

PROFESSIONAL SUMMARY

Backend Java Developer with 1+ year of experience building and optimizing scalable applications using Java and Spring Boot. Hands on experience implementing multithreading logic in production systems to improve performance and concurrency. Familiar with AWS and Azure cloud platforms, CI/CD pipelines, and 3-tier architecture deployments. Contributed to applied AI systems through prompt tuning using Gemini models. AWS Certified Solutions Architect – Associate and Cloud Practitioner.

WORK EXPERIENCE

Backend Engineer, Cognizant

01/2025 – Present

Ai Powered Proctoring platform(Leading US online Exam provider)

Backend Engineer

[Tech] Java, Spring Boot, Azure SDK, Gemini LLM , Microservices

[GenAI] Gemini-based multimodal inference pipeline that converts image + prompt inputs into structured JSON responses using schema-driven output generation, enabling type-safe deserialization, asynchronous processing, and scalable proctoring analysis.

[Concurrency/Scalability] Redesigned the Gemini-Flash-2.5 inference processing to use Java Virtual Thread pool instead of traditional thread pools. This eliminated thread starvation for I/O operations and reduced memory overhead from ~2MB to ~1KB per thread and reduced the chance of throttling and Azure service Bus queue and enhanced concurrency.

[Maintainability with SOLID] Refactored Gemini inference to an Open/Closed principle-compliant strategy architecture, allowing new execution behaviours to be added without changing the core service file.

[Azure SDK] Engineered asynchronous job processing pipeline with the help of Azure Service Bus with self-scheduling pattern for continuous frame extraction from video streams enabling the parallel processing up to 3 concurrent messages.

[Version Control] Participated in code reviews to ensure coding standards and best practices.

[Prompt Engineering] Automated prompt tuning across 1000+ real test images scenarios and enhanced LLM accuracy by 0.7% while reducing input token usage and operational costs.

PROJECTS

Three Tier Architecture Deployment

AWS EC2, RDS, VPC, ALB, Terraform, GitHub Actions, Networking, Linux

- Designed and deployed a scalable 3-tier architecture on AWS using EC2, RDS (MySQL), and VPC networking.

- Configured public/private subnets, NAT Gateway, security groups, and route tables for secure traffic flow.

- Implemented Auto Scaling Groups and Application Load Balancer for high availability, with RDS Multi-AZ and read replicas.

- Automated instance setup via user data scripts and managed services with systemd.

- Integrated CI/CD pipeline using GitHub Actions to automate deployment workflows, enabling consistent and repeatable infrastructure updates.

- Provisioned and managed AWS infrastructure as code using Terraform, enabling version-controlled, repeatable, and automated resource provisioning across environments.

EDUCATION

B.Tech Computer Science, GMR Institute of Technology ,Andhra Pradesh

2021 – 2025

8.8 CGPA

SKILLS

Languages: Java, Python

Frameworks: Spring , Spring Boot

Cloud&Devops: AWS: EC2, RDS, IAM, VPC , ALB | Azure: Azure Blob Storage, Azure Service Bus

Other Operating Systems: Linux

Tools: Git,GitHub,GitHub Actions,Terraform(IaC),Docker

API Testing: TestFully / Postman / SwaggerUI

Database: MySQL

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-Co3) verifyBadge (issued Feb'2026)

AWS Certified Cloud Practitioner (CCP) verifyBadge (Issued Sep'2025 )